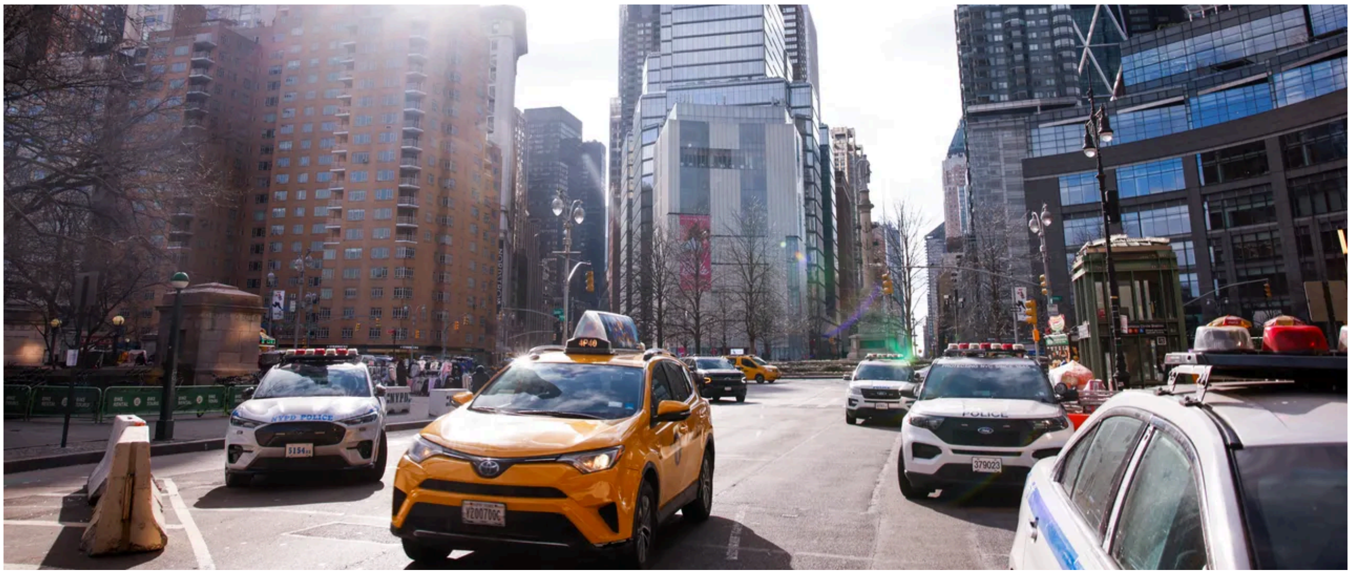
MATT BURGESS DHRUV MEHROTRA SECURITY JAN 7, 2025 1:30 PM

License Plate Readers Are Leaking Real-Time Video Feeds and Vehicle Data

Misconfigured license-plate-recognition systems reveal the livestreams of individual cameras and the wealth of data they collect about every vehicle that passes by them.



PHOTOGRAPH: KENA BETANCUR/GETTY IMAGES

IN JUST 20 minutes this morning, an automated license-plate-recognition (ALPR) system with an IP address geolocated to Tennessee captured photographs and detailed information from nearly 1,000 vehicles as they passed by. Among them: eight black Jeep Wranglers, six Honda Accords, an ambulance, and a yellow Ford Fiesta with a vanity plate.

This trove of real-time vehicle data, collected by one of Motorola's ALPR systems, is meant to be accessible by law enforcement. However, a flaw discovered by a security researcher has exposed live video feeds and detailed records of passing vehicles, revealing the staggering scale of surveillance enabled by this widespread technology.

More than 150 Motorola ALPR cameras have exposed their video feeds and leaking data in recent months, according to security researcher Matt Brown, who first publicized the issues in a series of YouTube videos after buying an ALPR camera on eBay and reverse engineering it.

As well as broadcasting live footage accessible to anyone on the internet, the misconfigured cameras also exposed data they have collected, including photos of cars and logs of license plates. The real-time video and data feeds don't require any usernames or passwords to access.

Alongside other technologists, WIRED has reviewed video feeds from several of the cameras, confirming vehicle data—including makes, models, and colors of cars—have been accidentally exposed. Motorola confirmed the exposures, telling WIRED it was working with its customers to close the access.

Over the last decade, thousands of ALPR cameras have appeared in towns and cities across the US. The cameras, which are manufactured by companies such as

“Every one of them that I found exposed was in a fixed location over some roadway,” Brown, who runs cybersecurity company Brown Fine Security, tells WIRED. The exposed video feeds each cover a single lane of traffic, with cars driving through the camera’s view. In some streams, snow is falling. Brown found two streams for each exposed camera system, one in color and another in infrared.

Broadly, when a car passes an ALPR camera, a photograph of the vehicle is taken, and the system uses machine learning to extract text from the license plate. This is stored alongside details such as where the photograph was taken, the time, as well as metadata such as the make and model of the vehicle.

Brown says the camera feeds and vehicle data were likely exposed as they had not been set up on private networks, possibly by law enforcement bodies deploying them, and instead exposed to the internet without any authentication. “It’s been misconfigured. It shouldn’t be open on the public internet,” he says.

WIRED tested the flaw by analyzing data streams from 37 different IP addresses apparently tied to Motorola cameras, spanning more than a dozen cities across the United States, from Omaha, Nebraska, to New York City. Within just 20 minutes, those cameras recorded the make, model, color, and license plates of nearly 4,000 vehicles. Some cars were even captured multiple times—up to three times in some cases—as they passed different cameras.

Jehan Wickramasuriya, corporate vice president overseeing license plate recognition products at Motorola Solutions, confirmed to WIRED that some devices were exposed, and that the company plans to introduce new security measures going forward.

being exposed.

“We are working directly with these customers to restore their system configurations consistent with our recommendations and industry best practices,” Wickramasuriya says. “Our next firmware update will introduce additional security hardening.”

“By leaving these incredibly insecure tracking devices on the open internet, police have not only breached public trust but created a bounty of location data for everyone who drives by which can be abused by stalkers and other criminals,” says Cooper Quintin, senior staff technologist at the Electronic Frontier Foundation, which last year found security vulnerabilities in ALPR cameras. “Police shouldn't be collecting this data at all unless there is an active investigation, and even then, the devices must be strictly scrutinized for security and public safety."

Brown initially found the exposed camera data after recently buying one of Motorola’s ALPR cameras on eBay, he says, and reverse engineered it to extract the device’s firmware. The researcher says he found details of both the color and infrared video streams on the device he purchased and was able to access the video from the camera in his testing lab.

Brown then set out to see whether any devices in the real world were publicly available online. Brown was able to use text from a “404” error page shown by the cameras—including unique language and peculiar grammar—to find IP addresses of exposed devices on the public internet. “I think that is a very unique type of error page that only exists on this device,” Brown says.

More than 150 results appear when using publicly available internet-scanning tools. The researcher says these likely belong to a sort of “hub” that is connected for such widespread surveillance systems.

“This is part of a general pattern where governments are inclined to roll out technical systems to meet the specific goals they have without thinking about, let alone working on, the potential negative impacts that those systems have and doing the work that you need to do to minimize the negative impacts,” says Daniel Kahn Gillmor, a senior staff technologist at the American Civil Liberties Union.

Gillmor points to New Hampshire’s ALPR law as one that is “reasonable.” The law says records from cameras should not “be recorded or transmitted anywhere and shall be purged from the system within 3 minutes of their capture.”

Update 9:45 am EST, January 10, 2025: An ALPR system's IP address reviewed by WIRED geolocated it to Nashville, Tennessee. However, a Motorola spokesperson says “Nashville's is not among the customer-modified network configurations that potentially exposed certain IP addresses.” Given the relative imprecision of IP-based geolocation data, we have removed the specific reference to Nashville.

SENIOR WRITER

Dhruv Mehrotra is an investigative data reporter at WIRED, where he uses technology to find, build, and analyze datasets for accountability journalism. His reporting has examined surveillance, policing, data brokers, platforms, and the government agencies and companies that use them. Before joining WIRED, he worked at Bloomberg News, the Center for ... Read More

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